

GRIFOLS Diagnostic Grifols, S.A.	Reference:	REGD-0000826	Version:	4.0, CURRENT
	Status:	Effective	Effective Date:	12-Jul-2011
Title:	SM DG57-DG517. Gripper and gripper pins replacement.			Page 2 of 9

DG-57 / DG-517 Service Manual

GRIPPER AND GRIPPER PINS REPLACEMENT

Procedure description
This procedure describes how to replace the gripper and pins. <i>Note: Document REGD-0000826_v4 replaces document REGD-0000826_v3.</i>

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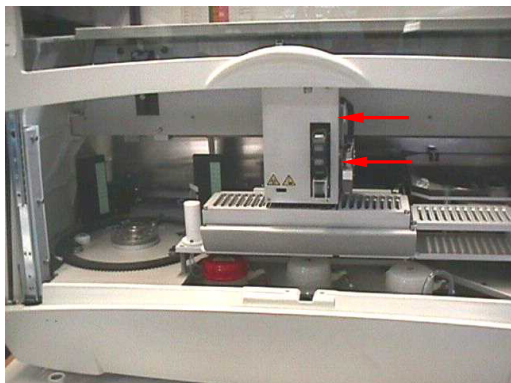
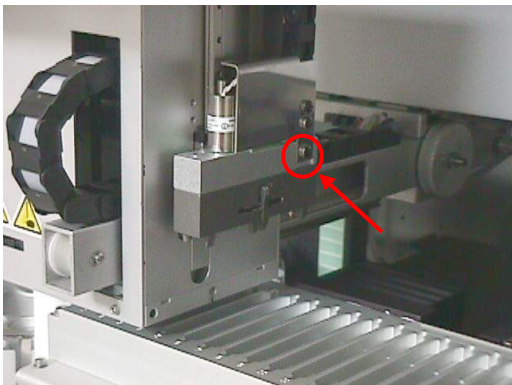
EQUIPMENT

1. An instrument connected to a PC with the software installed.
2. 232245 GRIPPER ADJUSTMENT TOOL (Ver.3)
3. Four new gripper pins (P/N: 232049).

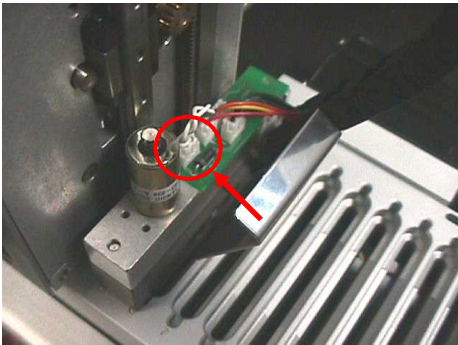
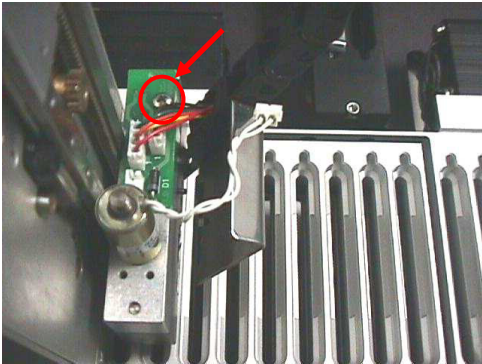
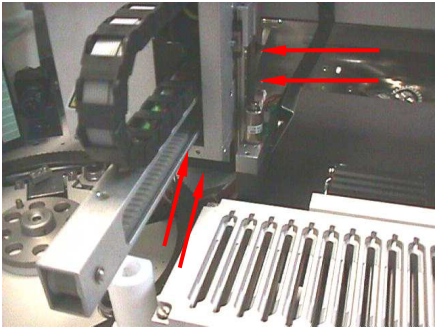
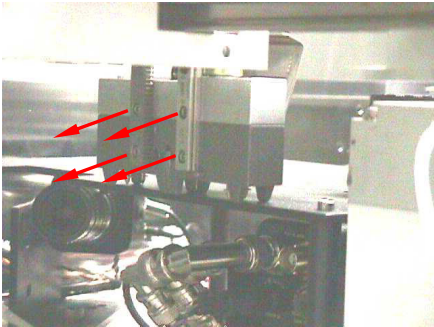
SET UP

1. Exit from the software if the software is running.
2. Turn off the instrument if it is turned on.
3. Open the door.
4. Extract the reagents carousel.
5. Extract the samples carousel.

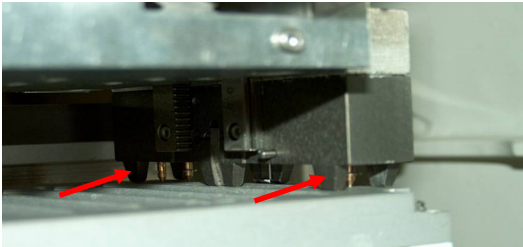
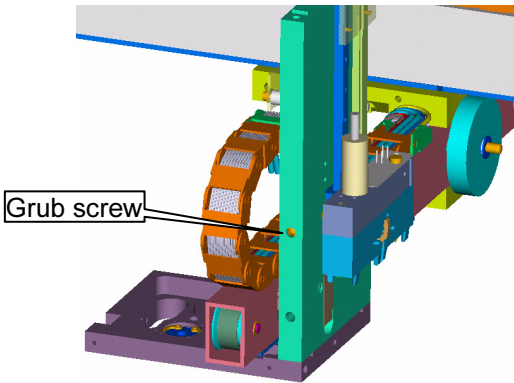
PROCEDURE

Process	Expected Results
1. Move manually the instrument arm to the maximum right position.  Picture 1	---
2. Unscrew the protection plate of the gripper.  Picture 2	---

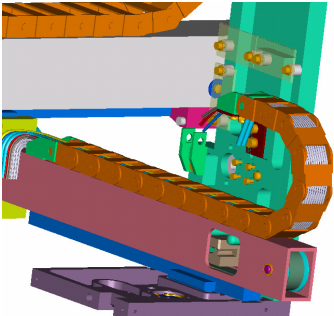
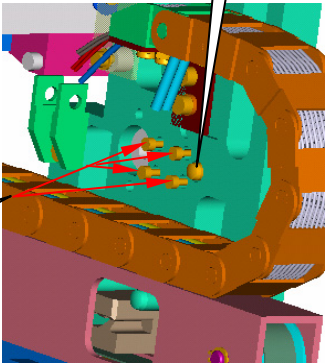
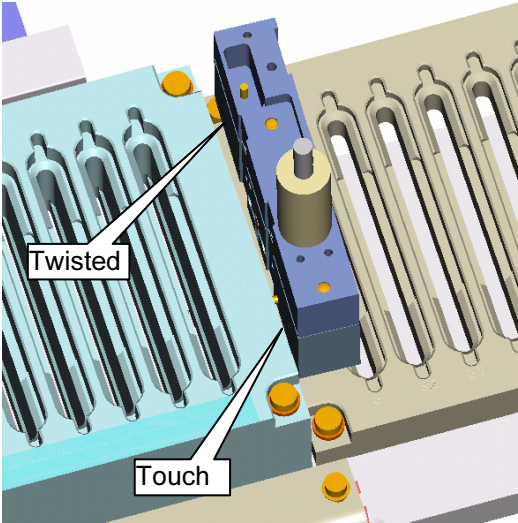
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Process	Expected Results
3. Disconnect the solenoid connector.  Picture 3	---
4. Unscrew the PCA and extract it from the gripper.  Picture 4	---
5. Locate, manually, the instrument arm to X axis maximum position and to the Y axis minimum position. Hold the gripper above the black protection of the reader. (See picture 5). Unscrew the 4 screws of the gripper.   Picture 5 Picture 6 Extract the gripper out of the instrument.	---

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Process	Expected Results
13. Screw PCA to the gripper. Reconnect solenoid cable. 14. Verify the vertical adjustment of the gripper, all the “legs” of the gripper must touch the incubator surface at the same time.  Picture 12 If the “legs” don’t touch the incubator surface at the same time, proceed to locate the gripper to the position described in step 5. Adjust the gripper position unscrewing the 4 screws of the gripper, positioning correctly the gripper touching the “legs” at the same time in the black protection of the reader. Screw again in a crossed way. 15. Check the vertical adjustment of the gripper repeating the first part of the step 12. If the gripper remains not correctly adjusted proceed to apply the next procedure: Remove the gripper motor following the REGD-0000824 procedure. Loose the four screws from the linear guide. Don’t remove them completely, just allow its movement vertically. Be careful not to remove the bolt from its position. If so, place it again. Tighten the grub screw, so it will locate the moving part of the linear guide to its ideal position. Tighten the four screws.  Picture 14	--- --- ---

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Process	Expected Results
  Picture 15 Picture 16 16. Verify the horizontal adjustment of the gripper, it means, the gripper has to be parallel to the Y-axis of the incubator. Locate, manually, the gripper to the position where the left surface of the gripper touches to the right surface of the incubator:  Picture 17 - If the gripper touches the incubator surface in its entire surface is not necessary any adjustment of the instrument arm and the procedure it's finished. - If the gripper appears twisted from the incubator surface is necessary to adjust the arm position.  Picture 18	---

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Process	Expected Results
17. The horizontal adjustment is done by moving the whole robot tower which is attached by 4 screws to a linear guide attached to Y-axis. - Locate, manually, the arm under the service rack. - Loose the grub screw. - Loose the four screws from the linear guide. Don't remove them completely, as the Z tower would fall down. - Tighten the grub screw, so it will locate the moving part of the linear guide to its ideal position. - Tighten the four screws.	---
18. Check the horizontal parallelism of the gripper (see step 14).	---
19. Adjust the instrument following the procedures REGD-0000807 and REGD-0000808.	---

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DOCUMENT CHANGE HISTORY

Document change history			
Version	Date	Change	Rationale
2.0	2005/05/09	<i>Document REGD-0000826_v2 replaces document 57053V0M000.</i>	Document reference changed
3.0	2011/04/28	Added 'previous version 'data', 'change' and 'rationale' inside document change history section.	Update change history for previous document history modification.
3.0	2011/04/28	Updated the Document Change History: - Date format changed to YYYY/MM/DD - Dates changed to document version effective date.	Adoption of ISO 8601 date format to avoid ambiguity.
3.0	2011/04/28	ID-GELSTATION and PROVUE erased from the title.	Service manual intended for Grifols market.
3.0	2011/04/28	<i>Procedure description : Document REGD-0000826_v3 replaces document REGD-0000826_v2.</i>	Document reference changed
3.0	2011/04/28	Modified spare part names and spare part code inside 'equipment' section.	Document change to update part numbers.
4.0	See document header.	<i>Document REGD-0000826_v4 replaces document REGD-0000826_v3.</i>	Document version changed.
4.0	See document header.	Document title.	-Commercial reference substituted by project reference. -Backslash replaced by words.
4.0	See document header.	Document header.	Increase information to easy document search.